

Magnesium

12
Mg

This lightweight metal is very reactive, including with oxygen in the air. When it burns, it produces a bright, white glow that is so intense, it can damage the eyes of anyone looking directly at the burning metal. Magnesium is used in flares and fireworks.



Yellowing leaves are a sign of magnesium deficiency.



Laboratory sample of pure magnesium



Finding magnesium

Magnesium and calcium were once thought to be the same element. In 1755, Scottish chemist Joseph Black (above) proved that they were different, but pure magnesium was only isolated in 1808 by Sir Humphry Davy.

Translucent crystals of magnesium carbonate



Brittle mineral

Magnesite

This mineral is mostly made up of magnesium carbonate, which is used in flooring, cosmetics, and even toothpaste. When heated strongly, it forms magnesium oxide, which is used to line furnaces due to its ability to resist heat.

These car rims are made of a magnesium-aluminum alloy for high strength and quality.



Custom-made car rims

Chlorophyll

Plants need sunlight to grow. Chlorophyll, a green-colored compound found in plants, absorbs light energy from the Sun, allowing plants to convert carbon dioxide and water into oxygen and sugars in a process called photosynthesis. At the center of each chlorophyll molecule is a magnesium atom.

Medication cup helps measure the required dosage.



Tummy settler

Magnesium carbonate is added to medication used to ease heartburn and other stomach ailments. The carbonate reacts with excess acid in the stomach, producing water, magnesium chloride, and carbon dioxide gas in the process. The gas can make you burp!

Magnesium alloys

Alloys containing magnesium have the advantage of being both strong and lightweight. For that reason, several of them are used in the manufacture of high-performance machinery. One such alloy is called Mag-Thor: a mixture of magnesium, thorium, and other elements, it is used to build aircraft engines.